

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Electrical Contact Cleaner, Part Number 3824510, or equivalent.

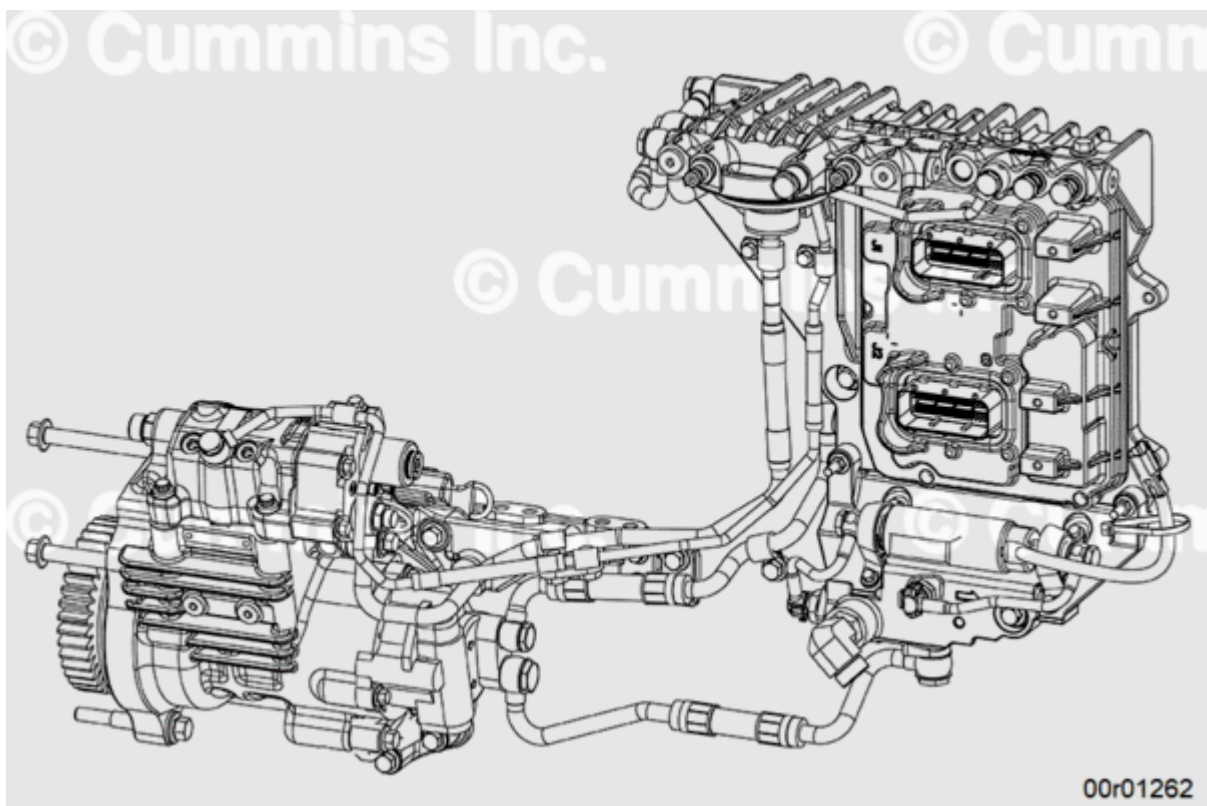
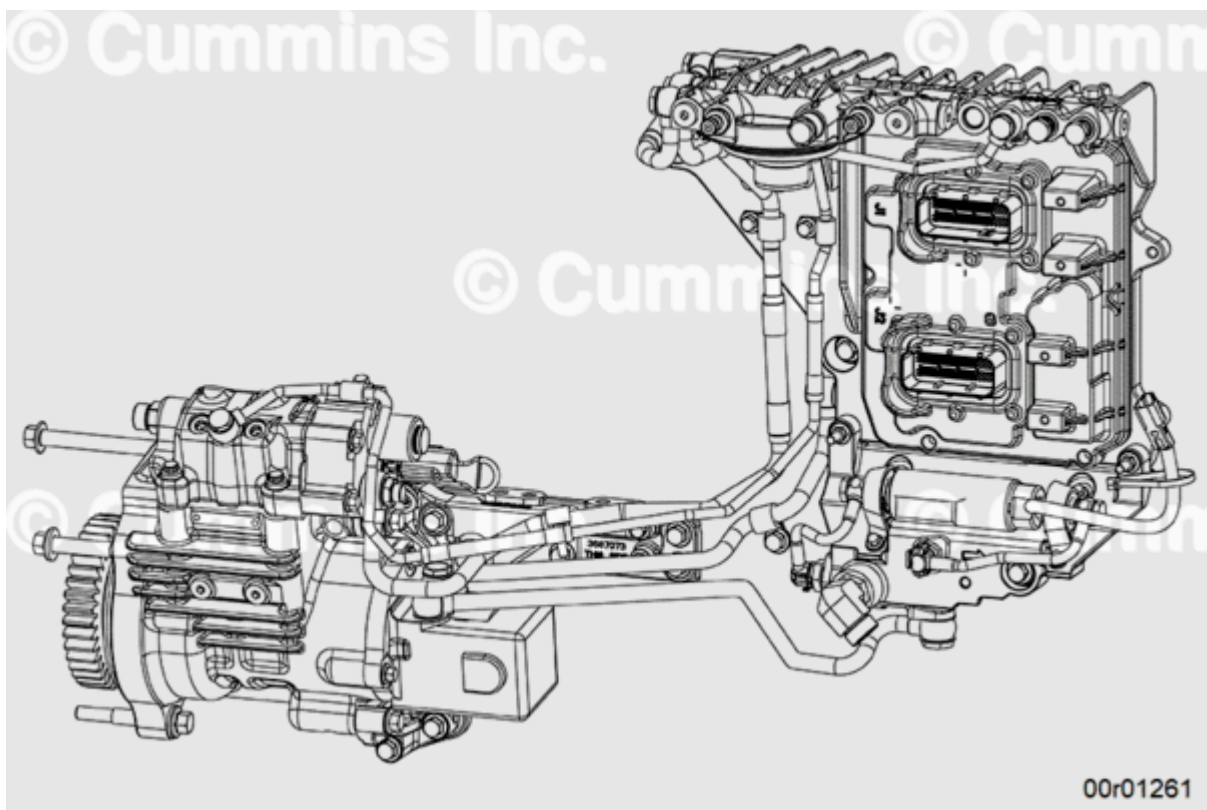
Additional Service Items

- No additional service items required.

General Information

Engines can be equipped with two fuel pump assembly variations. These fuel pumps will have the same performance despite gear pump visual variation(s). Each fuel pump requires distinct fuel line configurations.

Cummins® Service Fuel Pumps:



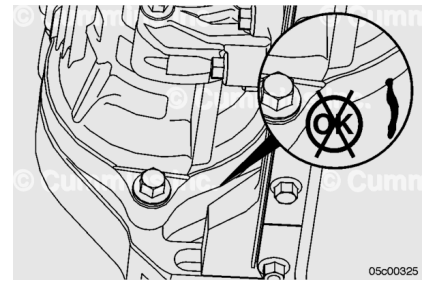
Initial Check

The gear pump shaft incorporates a double seal. The first one seals oil in the fuel pump cam housing; the second one seals fuel in the gear pump. If either seal malfunctions, fluid will drip from the weep hole.

The weep hole is located near the gear pump mounting flange on the inboard side (toward the engine block).

A gear pump o-ring seals oil at the gear pump to cam housing interface.

Use the following procedure for details on measuring the fuel pump gear pump output pressure. Refer to Procedure 005-236 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-236.html>)



Preparatory Steps

⚠ WARNING ⚠

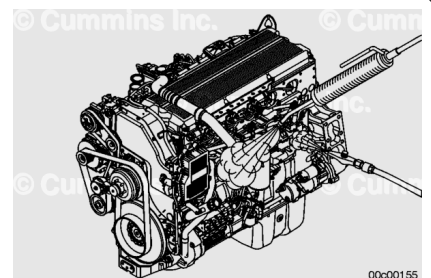
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.



Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.) which would expose the fuel system, or internal engine component, to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system

and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

(/qs3/pubsys2/xml/en/procedures/377/377-000-009.html)

To prevent engine damage from debris or contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ WARNING ⚠

Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.

⚠ CAUTION ⚠

Use caution when disconnecting or removing fuel lines, replacing filters and priming the fuel system that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be discarded in accordance with local environmental regulations.

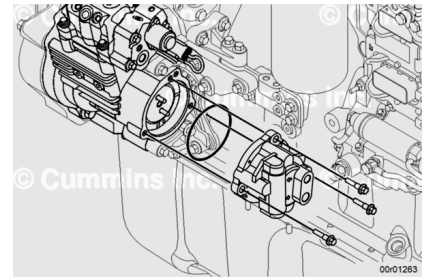
- Disconnect the batteries. See equipment manufacturer service information.
- Clean the fuel, oil, and debris from the gear pump.
- Remove the fuel supply lines from the gear pump. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-024.html)

Remove

Remove the three capscrews that hold the gear pump to the fuel pump.

Remove the gear pump from the fuel pump.

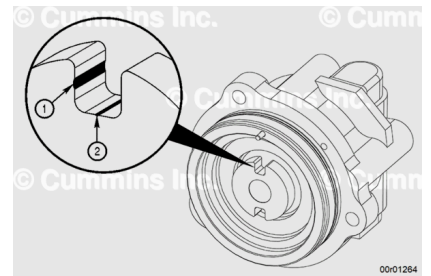
Remove the gear pump o-ring.



Inspect for Reuse

If the gear pump shaft is broken, inspect the fuel pump gear pump coupling for wear at the drive notch.

- Wear on the sides of the drive notch (1) is normal.
- Wear on the bottom of the drive notch (2) may indicate journal bearing wear in the high-pressure fuel pump.

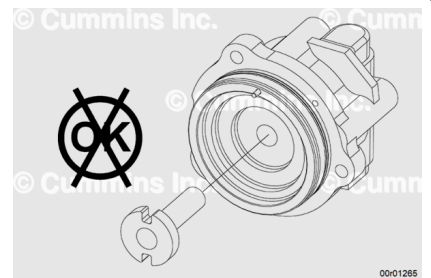


If the gear pump shaft can be pulled out by hand either:

- The shaft broke because the gear seized in the gear pump
- Or, the bushings are spun in the high-pressure pump.

If damage is found, inspect the high-pressure fuel pump bushing(s) and internal components. Use the following procedures for additional information.

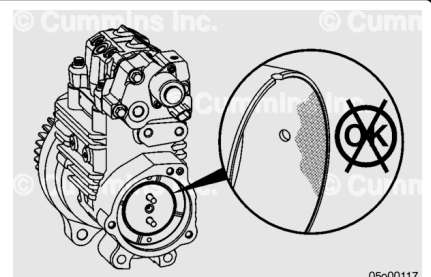
- Refer to Procedure 005-016 in Section 5.
(</qs3/pubsys2/xml/en/procedures/377/377-005-016.html>)
- Refer to Procedure 005-227 in Section 5.
(</qs3/pubsys2/xml/en/procedures/377/377-005-227.html>)



With the fuel gear pump removed from the high-pressure pump, inspect the rear high-pressure pump bushing.

Replace the high-pressure fuel pump if either the front or rear bushing has malfunctioned.

- Bushing support shown signs of excessive heat
- Bushing is melted or damaged.



Install

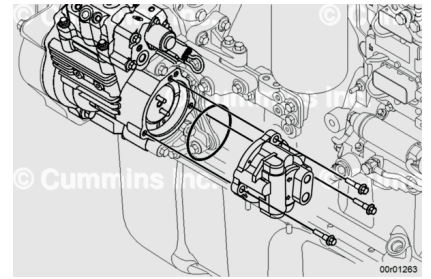
Install the o-ring onto the fuel pump gear pump.

Insert the mounting bolts through the gear pump flange.

Index the gear pump input shaft to engage the drive pins and install the gear pump.

Install the three gear pump capscrews and tighten.

Torque Value: 25 n•m [221 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Hold the banjo fittings while tightening the banjo bolt to prevent fitting rotation. Allowing the banjo fitting to rotate can damage the fuel line.

- Install the fuel supply lines. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-024.html>)
- Connect the battery. See equipment manufacturer service information.
- Operate the engine and check for leaks.